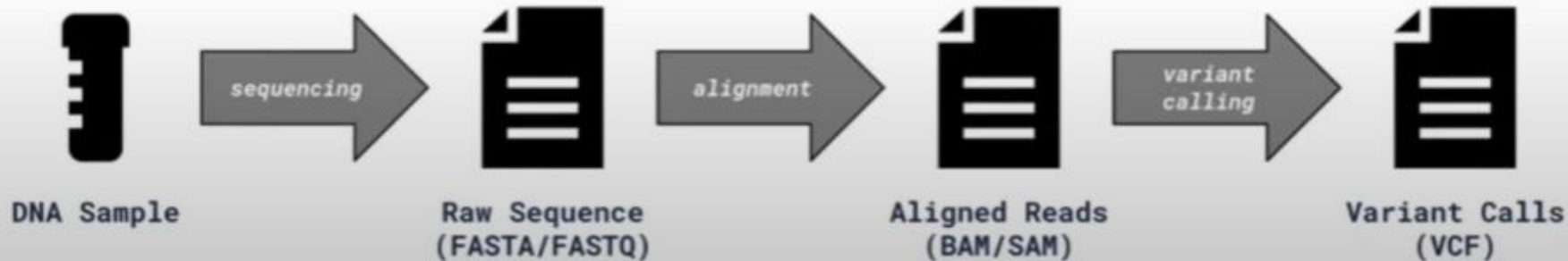


GATK4 WORKFLOW SOP

(fastq to vcf)



List of files:

- 1) Fastq files (R1 & R2)
- 2) Reference file (hg37/hg38)

List of tools (Install latest version):

- **Fastp**
- **Multic**
- **BWA**
- **Samtools**
- **Picard**
- **GATK4**

STEP1: GATK4- Concatenating datasets & trimming with fastp:

fastp installation: Open terminal -> Download master .zip file from the Github link (wget <https://github.com/OpenGene/fastp/archive/refs/tags/v0.20.1.tar.gz>) -> Unzip the file -> Open fastp-master from the folder -> Run install command : **sudo apt install fastp**

Run fastp:

To merge the R1 and R2 files in a single read file use the cat command:

```
cat name_R1.fastq.gz name_R2.fastq.gz > merged_R1_&_R2.fastq.gz
```

```
fastp -i *.fastq.gz -o fastpoutput_R1_&_R2.fastq.gz -h fastq.html -j fastq.json
```

#USE THE BELOW COMMAND FOR NON-MERGED R1 AND R2 FILES:

```
fastp -i *R1.fastq.gz -I *R2.fastq.gz -o name_outfastp_R1.fastq.gz -O  
name_8053055_outfastp_R2.fastq.gz
```

Output files: fastpoutput.fastq.gz, fastq.html, fastq.json

STEP2: GATK4 – Quality checks with multiqc:

Multiqc installation: Open terminal -> pip install multiqc -> Run the multiqc

Multiqc input file will be the .json file which was generated from the previous fastp step.

multiqc /path/to/.json/MULTIQCOUTPUT/

Output files: multiqc_report.html

STEP3: GATK4 – Mapping with BWA-MEM:

BWA installation: Open terminal -> wget

<https://sourceforge.net/projects/bio-bwa/files/latest/download> -> sudo apt install bwa mem

We need to index the reference file to run the BWA-MEM:

bwa indexing command:

bwa index reference.fna #IT WILL TAKE MORE TIME TO COMPLETE

Output files: .amb, .anm, .bwt, .pac, .sa

Once the indexing is completed, run the BWA-MEM command:

bwa mem reference.fna fastpoutput_R1_&_R2.fastq.gz > output_mapped.sam

Output files: output_mapped.sam (.sam file)

STEP4: Converting SAM file to BAM with Samtools:

Samtools installation: Open terminal -> wget

<https://github.com/samtools/samtools/releases/download/1.17/samtools-1.17.tar.bz2>

-> sudo apt install samtools

1) Converting SAM file to BAM:

samtools view -S -b output_mapped.sam > output_mapped.bam

2) Converting unmapped BAM:

samtools view -f4 output_mapped.bam > output_unmapped.bam

Output files: output_unmapped.bam (.bam file)

STEP5: AddReplaceReadGroup, Sorting & MarkDuplicates with Picard:

Picard installation: Download the picard.jar file from this link

<https://github.com/broadinstitute/picard/releases/tag/3.0.0> -> java -jar picard.jar

Note: Always keep the picard.jar file inside the working directory.

1) AddOrReplaceReadGroups:

java -jar picard.jar AddOrReplaceReadGroups -I output_unmapped.bam -O output_picard.bam -RGLB lib1 -RGPL ILLUMINA -RGPU unit1 -RGSM 20

Output files: output_picard.bam (.bam file)

2) Sorting:

```
java -jar picard.jar SortSam \
```

```
I=output_picard.bam \
```

```
O=output_picard_sorted.bam \
```

```
SORT_ORDER=coordinate
```

Output files: picard_output_sorted.bam (sorted.bam and .bai file)

3) Mark duplication (marking and removing the duplicate reads dedup.bam file as output):

```
java -jar picard.jar MarkDuplicates \
```

```
I=output_picard_sorted.bam \
```

```
O=output_picard_sorted_dedup.bam \
```

```
M=marked_dup_metrics.txt
```

Output files: output_picard_sorted_dedup.bam (dedup.bam)

STEP6: GATK4- Variant Calling with Haplotypecaller:

GATK4 installation: Download the GATK4 zipped folder from this link <https://github.com/broadinstitute/gatk/releases> -> unzip the folder -> check whether the .jar files are present (gatk-package-[version]-local.jar,gatk-package-[version]-spark.jar)

Note: Always keep the .jar file inside the working directory or use
`export PATH="/path/to/gatk-directory:$PATH"`

GATK variant calling steps needs index file and dictionary files (.fai, .dict)

1) Index file command using samtools:

samtools faidx reference.fna

Output files: reference.fna.fai (.fai file)

2) Dictionary file command using picard:

**java -jar picard.jar CreateSequenceDictionary -REFERENCE reference.fna
-OUTPUT reference.dict**

Output files: reference.dict (.dict file)

After the index and dictionary files generation keep the 3 files (.fna .fai .dict) to the installed gatk folder.

3) Indexing the .bam file:

samtools index /path/to/dedup.bam/output_picard_sorted_dedup.bam

Output files: output_picard_sorted_dedup.bam (indexed dedup.bam file)

4) Run GATK4 Haplotypecaller:

```
java -jar gatk-package-4.4.0.0-local.jar HaplotypeCaller \
```

```
-R reference.fna \
```

```
-I output_picard_sorted_dedup.bam \
```

```
-O variants.vcf
```

Output files: variants.vcf (.vcf file)